

Diffusive surface scattering and hydrodynamics

This set of notes is on the following papers: Role of diffusive surface scattering in nonlocal plasmonics (<https://iopscience.iop.org/article/10.1088/1361-648X/ab977d/meta>) by **Svendsen et al** (which will be referred to as **Reference 1** below) and A generalized non-local optical response theory for plasmonic nanostructures (<https://www.nature.com/articles/ncomms4809>) by **Mortensen et al** (which will be referred to as **Reference 2** below). We ultimately wish to derive **Equation 5** and **Equation 6** of **Reference 1**, which we do in **Section 1** below. Edit <https://hackmd.io/@aligho/HJvv6hASGe/edit?both>

Section 1: Hydrodynamic approach with a diffusion constant

We start with **Equation 3** and **Equation 4** of **Reference 1**, which we restate below:

$$\partial_t \mathbf{v} = -\gamma \mathbf{v} - \frac{e}{m} \mathbf{E} - \frac{\beta^2}{n_0} \nabla n_1,$$

$$\partial_t n_1 + \nabla \cdot (n_0 \mathbf{v}) = D \nabla^2 n_1 \rightarrow \partial_t n_1 + \nabla \cdot [n_0 \mathbf{v} - D \nabla n_1] = 0,$$

from which we may easily read off Fick's current density, $\mathbf{J}^n$, from the quantity in brackets above. Unlike **Reference 1**, we define two different currents. $\mathbf{J}^n$ will be the number current and $\mathbf{J} = -e\mathbf{J}^n$ will be the charge current in the derivations below. We rearrange terms in the the force equation and take a time derivative of both sides as follows:

$$n_0 \partial_t \mathbf{v} + \gamma n_0 \mathbf{v} = -\frac{en_0}{m} \mathbf{E} - \beta^2 \nabla n_1 \rightarrow$$

$$\partial_t^2 \mathbf{J}^n + \gamma \partial_t \mathbf{J}^n + \partial_t^2 D \nabla n_1 + \gamma D \nabla \partial_t n_1 = -\frac{en_0}{m} \partial_t \mathbf{E} - \beta^2 \nabla \partial_t n_1,$$

where we also did a simple replacement: $n_0 \mathbf{v} \rightarrow D \nabla n_1 + \mathbf{J}^n$. We now take the time dependencies to be harmonic: $\propto e^{-i\omega t}$, which gives us the following relation in the frequency domain:

$$-\omega^2 \mathbf{J}^n - i\gamma \omega \mathbf{J}^n = \frac{ien_0 \omega}{m} \mathbf{E} + (\beta^2 + \gamma D) \nabla \nabla \cdot \mathbf{J}^n - i\omega D \nabla \nabla \cdot \mathbf{J}^n,$$

where we also used the charge conservation equation (**Equation 4**). We now re-express this equation in terms of the charge current, $\mathbf{J}^e \equiv -e \mathbf{J}^n$, giving us the following:

$$\mathbf{J}^e + \frac{1}{\omega^2 + i\omega\gamma} \left[\beta^2 + \gamma D - i\omega D \right] \nabla \nabla \cdot \mathbf{J}^e = \frac{ie^2 n_0}{m(\omega + i\gamma)},$$

which is **Equation 5** and **Equation 6**. From this, we see that augmenting the usual hydrodynamic approach means making the following substitution:

$$(\beta/c)^2 \rightarrow (\beta/c)^2 + (\gamma/\omega_p)(D\omega_p/c^2) - i(\omega/\omega_p)(D\omega_p/c^2)$$

$$\text{We take } D \approx v_F^2/\gamma \rightarrow D\omega_p/c^2 \approx (v_F/c)^2(\omega_p/\gamma)$$